AliyunConsoleAgent: Training Web Agents in Real-World Cloud Environments via Distillation and Reinforcement Learning

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Paper version: arXiv 2606.09447v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Console UIs drift from their docs; manual coverage is below 1%.*

Execution-as-Verification: an agent runs each documented procedure on the live console.

Motivation: three obstacles to a private agent

1. **Scale.** About 4 million inspections a year.
2. **Cost.** Frontier APIs: about CNY 7 per task.
3. **Noisy RL.** Missing prerequisites look like agent errors.

Evaluation setup

278 tasks · 12 products | 76 standard · 202 hard | Qwen3-VL-32B base | ActionTrail rule check | 3 runs per task | 40 turns per sub-step

- **Online:** live console. Single-step accuracy is offline.
- **Held out:** no benchmark document URL appears in SFT or RL.

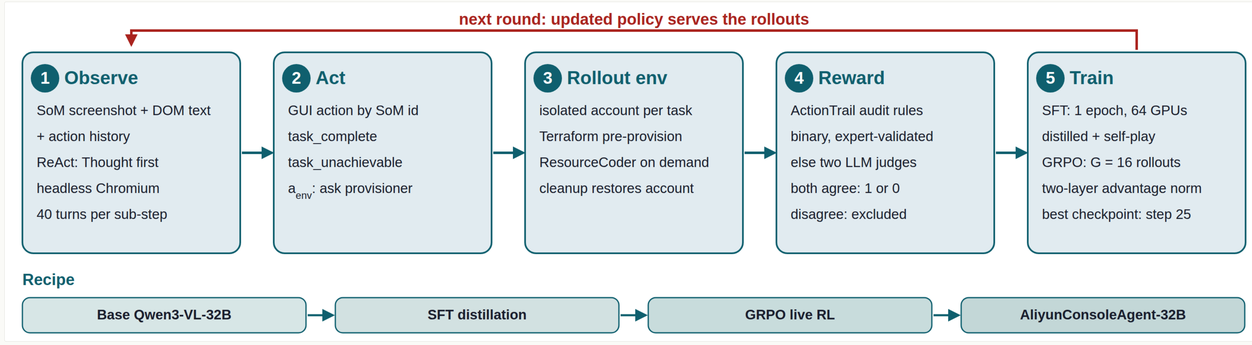
Our read (CUA Radar)

69	★★	online
score of 100	band	evidence level

- **Reports:** online eval; 3 runs; baselines; cost; ablation; reward check; code.
- **Partial:** held-out by document; budget per sub-step.
- Pattern K12. Authors' numbers, not rerun.

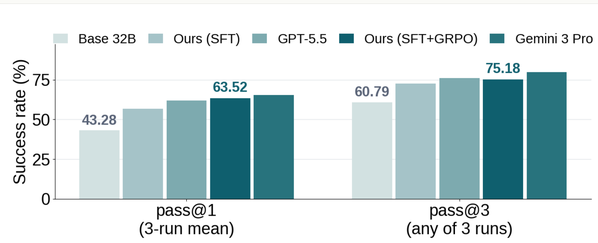
Method: SFT on distilled trajectories, then GRPO on live accounts

- **Sees / acts:** SoM screenshot, DOM text, history in; element-id GUI action or provisioning call out.
- **Training:** about 160K single-step SFT samples (distilled and self-explored), then GRPO with G = 16 rollouts.
- **Signal:** ActionTrail audit rules; else two LLM judges that must agree, and a disagreement drops the trajectory.



Our redraw (not the paper's figure): the agent loop, rollout environment and SFT-then-GRPO recipe.

Key results: success on 278 live tasks (%)



Tables 3 and 4, redrawn: pass@1 is a 3-run mean.

63.52% pass@1 vs 65.34% Gemini 3 Pro; CI [−1.27, 7.39]

SO WHAT → *Cost per task: Gemini 3 Pro 7.00 CNY, ours 0.56. Also in Table 3: Kimi K2.6 60.79%, Qwen3.6-Plus 57.69% pass@1. On the 202 hard tasks, pass@3 is 71.78% (ours) vs 76.73% (Gemini).*

Ablation: training stages, pass@1 (%)

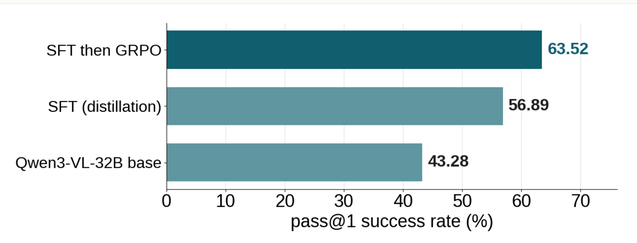


Table 5, redrawn: post-SFT strategies on the 278 tasks.

SO WHAT → *SFT adds +13.61 pp; GRPO adds +6.63 pp more, 95% CI [2.97, 9.64]. Table 5 has only these three rows: no RL-only or longer-SFT control is reported. GRPO checkpoint chosen by validation every 5 steps (best at step 25). Per task, 9 move from SFT failure to GRPO success; none falls from 3/3 to 0/3. The authors credit online exploration.*

Headline Numbers

63.52%
pass@1, 278 live tasks
3-run mean

- 65.34% Gemini 3 Pro
- 92% lower cost
- 75.18% pass@3

The paper's own controls

CONTROL	OURS (32B)	COMPARED WITH
pass@1 std, 3 runs	SFT+GRPO ±1.86	Gemini ±2.15
95% CI, GRPO vs SFT	[2.97, 9.64]	Gemini gap [−1.27, 7.39]
pass@3 standard / hard	84.21 / 71.78%	Gemini 88.16 / 76.73%
Offline single-step	92.75% (SFT)	Gemini 95.25%

Takeaway

SFT then GRPO lifts a 32B model from 43.28% to 63.52% on live tasks.

0.56 CNY per task | pass@3: 75.18% | ~300 tasks/hour

Scan to Read

Paper | Code